

Types of cognitive load

There are three types of cognitive load, namely, intrinsic, extraneous and germane, as shown in Fig. 2.

Intrinsic load is defined as the basic cognitive load required for performing a task. It is directly proportional to the difficulty of the task and is inversely proportional to the degree of expertise of the performer.

Extraneous load is defined as the wasted cognitive load that does not relate to the primary cognitive learning activities but emerges reluctantly. It is caused by factors that are not central to the material to be learned, such as presentation methods or activities that require split attention between multiple sources of information. Instructional methods should avoid overloading

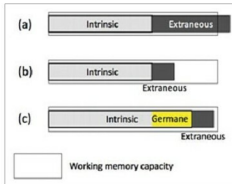


Fig. 2: Types of cognitive load

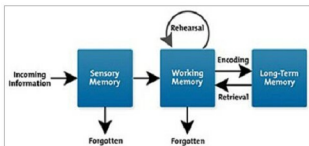


Fig. 3(a): Less extraneous load

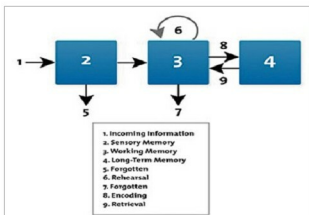


Fig. 3(b): High extraneous load

it with additional activities that do not directly contribute to learning.

As shown in Figs 3(a) and 3(b), there are two ways to present the same information to the learner, and one can clearly distinguish that, if the learner is presented with information as shown in Fig. 3(b), she or he has to put extra extraneous load to grasp the information as compared to a situation where information is presented in the form of Fig. 3(a).

Germane load is defined as the load used for learning, such as for constructing schema activities. These elements help facilitate the development of a learner's knowledge base.

All these loads should be within the working memory, and Fig. 2(a) shows that cognitive load exceeds the limits of the learner's working memory capacity due to an increase in extraneous load.

Fig. 2(b) shows that cognitive loads fall within a range, and in this scenario, a learner will perform a task easily and will show good results. In Fig. 2(c), cognitive loads are within range and some memory capacity is left spare and intrinsic load is transformed into germane load to foster the learning activities and process. Thus, Fig. 2(c) represents the ideal state of learning.

Now, the challenge is to measure cognitive load of the learner while she or he is learning, and the solution is an electroencephalogram (EEG) device.

EEG device: A tool to collect data about cognitive load

In the human brain, each individual neuron communicates with the others by sending tiny electrochemical signals. When millions of neurons are activated, each contributing with small electrical currents, these generate a signal that is strong enough to be detected by an EEG device. Recent availability of simple, low-cost and portable EEG monitoring devices makes it feasible to take this technology from the lab into schools/colleges.

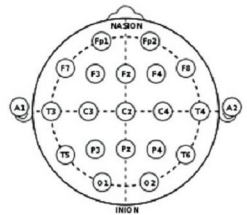


Fig. 4(a): International 10-20 system

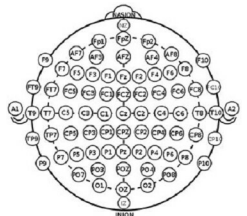


Fig. 4(b): Modified combinatorial nomenclature

EEG devices collect data from the human skull using electrodes that are placed at a number of positions. The positions to measure EEG signals are defined by International 10-20 system as shown in Fig. 4(a). It is an internationally-recognised method to describe and apply the location of scalp electrodes in the context of an EEG test or experiment. This system is based on the relationship between the location of an electrode and the underlying area of cerebral cortex. The 10 and 20 refer to the fact that actual distances between adjacent electrodes are either 10 or 20 percent of the total front-back or right-left distance of the skull.

A variant of this standard is known as Modified Combinatorial Nomenclature where data collection points are more, as shown in Fig. 4(b).

Brain electrical activity signals fall in the range of 1Hz to 32Hz, and are divided into four bands. Researchers have studied the correlation of these frequency band signals with the cognitive state of the user. Details of these frequency bands and their significance is given in Table 1.